PCIe x4 a 4x NVMe M.2

Ver: 1.0

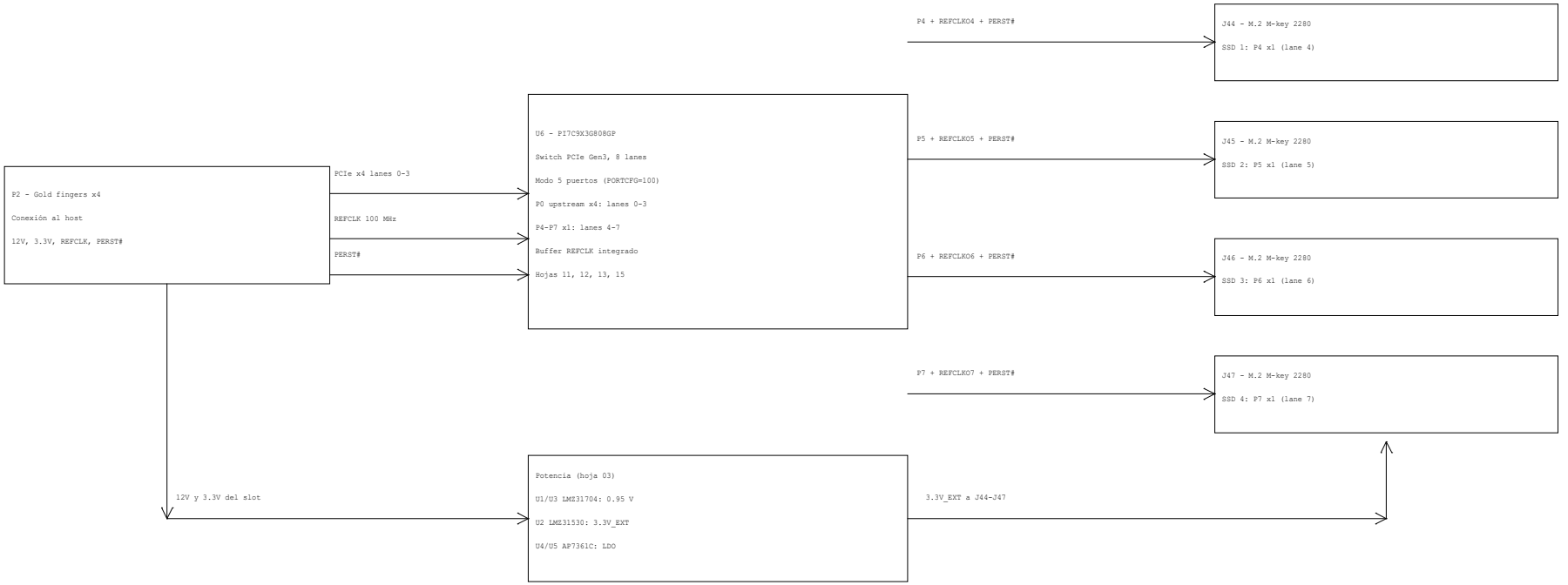
Titulo	Hoja
01_PORTADA	1
02_DIAGRAMA DE BLOQUES	2
03_ARBOL DE POTENCIA	3
04_RELOJ DE REFERENCIA	4
05_MAPEO DE PUERTOS	5
06_REGULADORES 3.3V Y 0.95V	6
07_LDO 1.8V Y 0.95V	7
08_UPSTREAM X4	8
10_CONECTORES M.2	9
11_PI7C9X3G808GP CONFIGURACION	10
12_PI7C9X3G808GP PARES DIFERENCIALES	11
13_PI7C9X3G808GP ALIMENTACION	12
15_BUFFER DE RELOJ	13
17_RESET PERST	14

Title

Size: Custom | Document Number | Rev: 1.6

Date: | Sheet 1 of 20

DIAGRAMA DE BLOQUES - PCIe Gen3 x4 a 4 SSD NVMe M.2

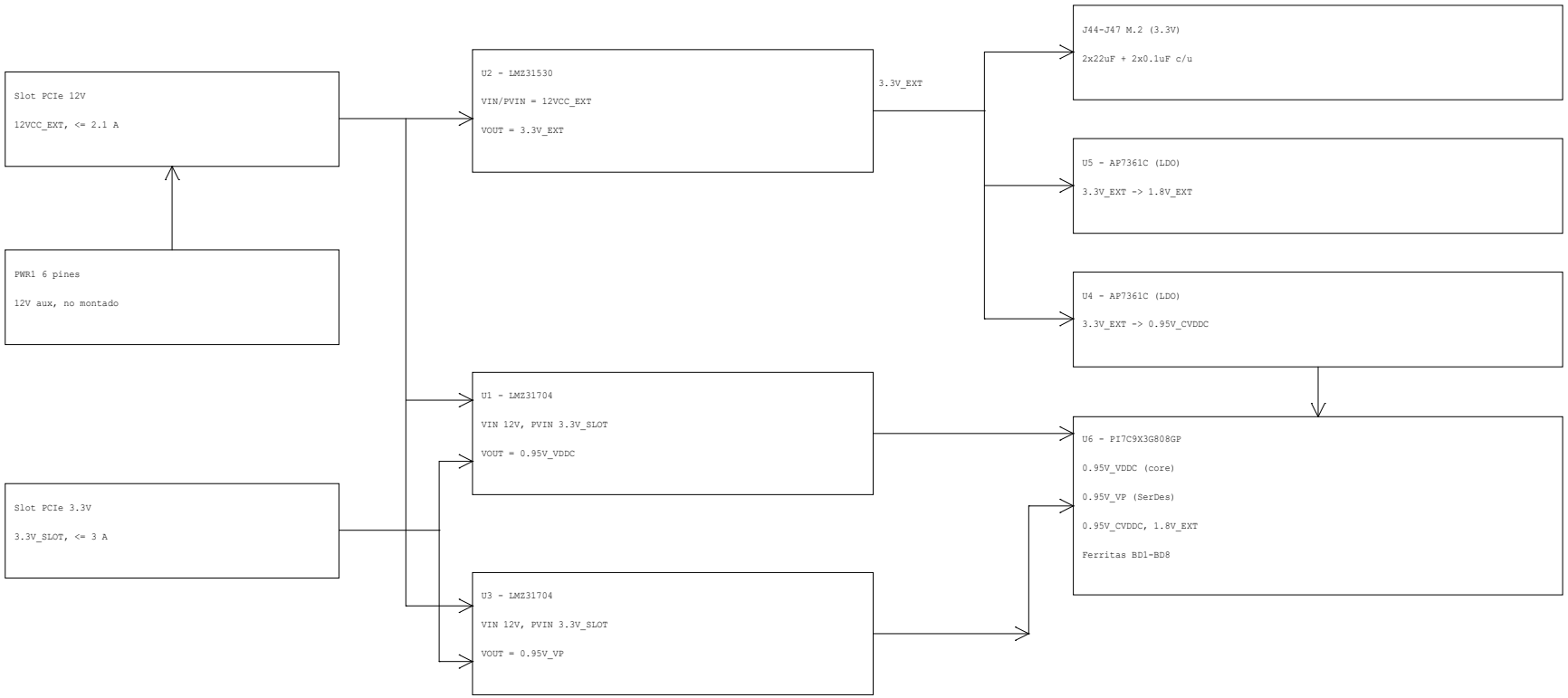


Cambios respecto al EVB Diodes V1.6: sin slots S1-S3, sin signal switches U12-U17, sin fuente ATX.

Conexion directa: gold fingers x4 -> U6; L4->J44, L5->J45, L6->J46, L7->J47.

Title: PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: C	Document Number:	Rev: 1.6
Date:	Sheet 2 of 20	

ÁRBOL DE POTENCIA

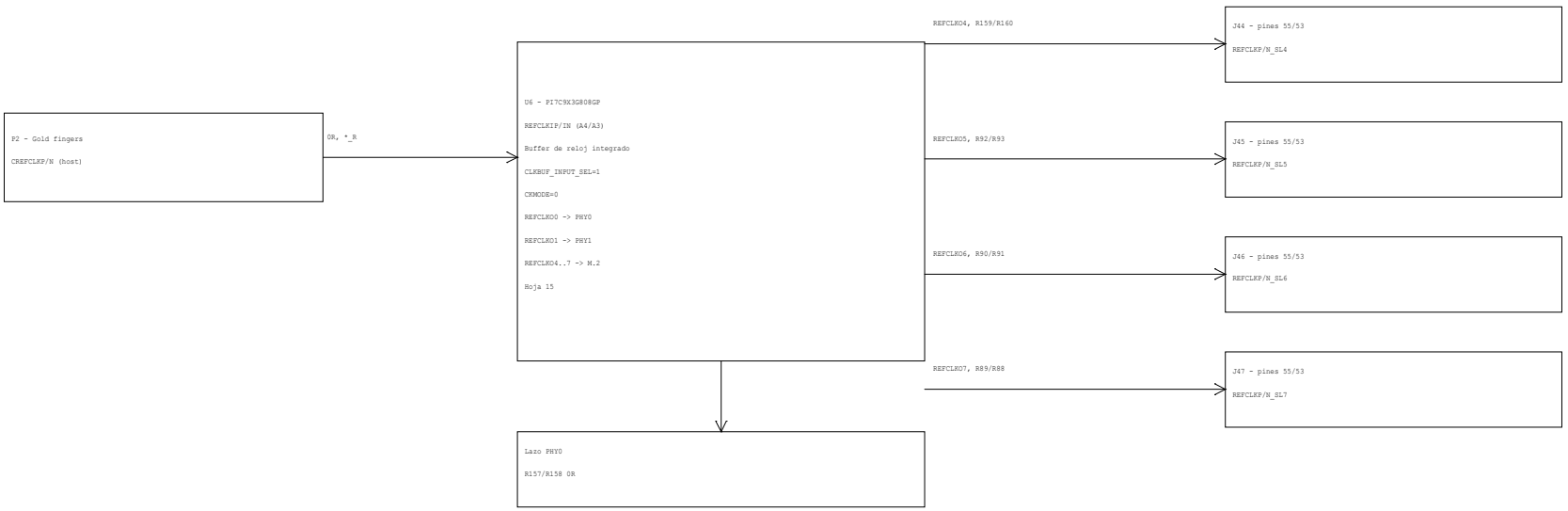


Presupuesto slot x4: 25 W (3.3V <= 3 A, 12V <= 2.1 A); 4 W por SSD.

0.95V_VDDC y 0.95V_VP en planos (TOP/ART04) en el layout.

Title: PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: Custom	Document Number	Rev: 1.6
Date:	Sheet 3 of 20	

RELOJ DE REFERENCIA (REFCLK) 100 MHz



Sin generador ni buffer externo (U20 del EVB eliminado).

Pares de 85 ohm diferencial, igualación dentro del par <= 5 mil.

Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number	Rev: 1.8
Date:	Sheet 4 of 20	

Mapeo de puertos y lanes

A. Selección del modo de funcionamiento

PORTCFG[2:0]

PORTCFG[2]	PORTCFG[1]	PORTCFG[0]	Modo de funcionamiento
0	0	1	2 puertos - 8 lanes
0	1	0	3 puertos - 8 lanes
0	1	1	4 puertos - 8 lanes
1	0	0	5 puertos - 8 lanes
1	0	1	8 puertos - 16 lanes

Chipmode[1:0]

Chipmode1	Chipmode0	Modo
0	0	Normal
0	1	Reservado
1	0	Reservado
1	1	Reservado

B. Este diseño: PORTCFG=100 (5 puertos), CHIPMODE=00

Lanes	Puerto	Conector	TX (C AC -> pin)	RX	REFCLK	PERST#
0-3	P0 x4 up	P2 fingers	PET0-3_P0	PER0-3	CREFCLK	CPERST_L
4	P4 x1	J44	C741/2 -> 47/49	43/41	55/53	50
5	P5 x1	J45	C751/2 -> 47/49	43/41	55/53	50
6	P6 x1	J46	C761/2 -> 47/49	43/41	55/53	50
7	P7 x1	J47	C771/2 -> 47/49	43/41	55/53	50

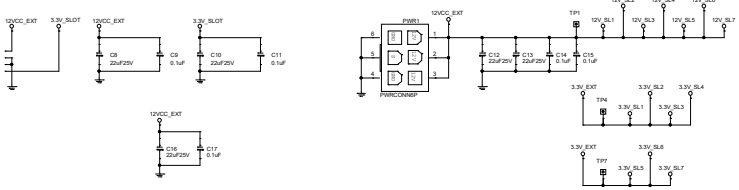
Sin signal switches, conexión directa.

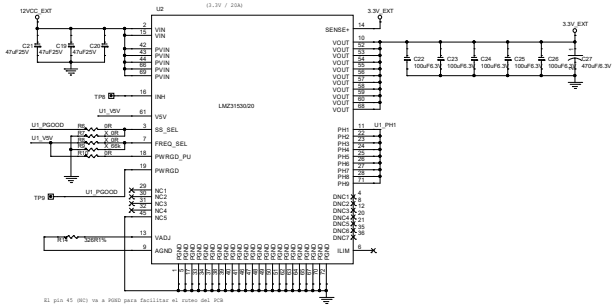
C. Puertos y lanes según el modo

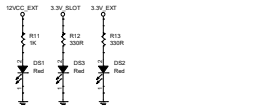
MODO	2 ptos	3 ptos	4 ptos	5 ptos	8 ptos
Lane 0	P0	P0	P0	P0	P0
Lane 1	P0	P0	P0	P0	P1
Lane 2	P0	P0	P1	P0	P2
Lane 3	P0	P0	P1	P0	P3
Lane 4	P4	P4	P4	P4	P4
Lane 5	P4	P4	P4	P5	P5
Lane 6	P4	P5	P5	P6	P6
Lane 7	P4	P5	P5	P7	P7

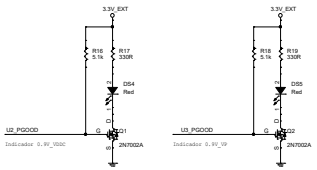
Title: PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: Custom	Document Number	Rev: 1.6
Date:	Sheet 5 of 20	

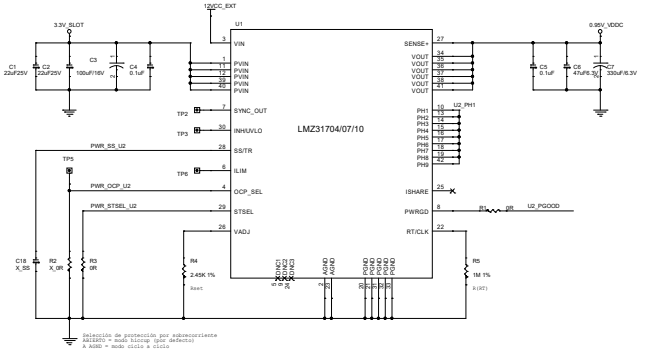
Conector 12V auxiliar (opcional)









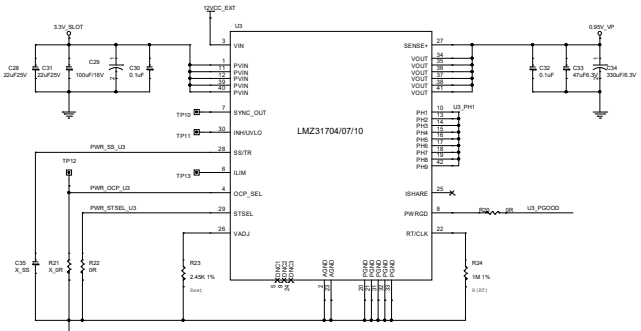


Capacitor y tiempo de arranque suave							
Css (nF)	ABIERTO	3.3	4.7	10	15	22	33
Tiempo SS (ms)	1.2	2.1	2.5	3.8	5.1	7.0	9.8

Configuración del módulo

Vout = [(1.43 / Rset) + 1] x 0.4

Vout	Rset	R(RT)	fsw
0.95V	2.45K	1M	250KHz



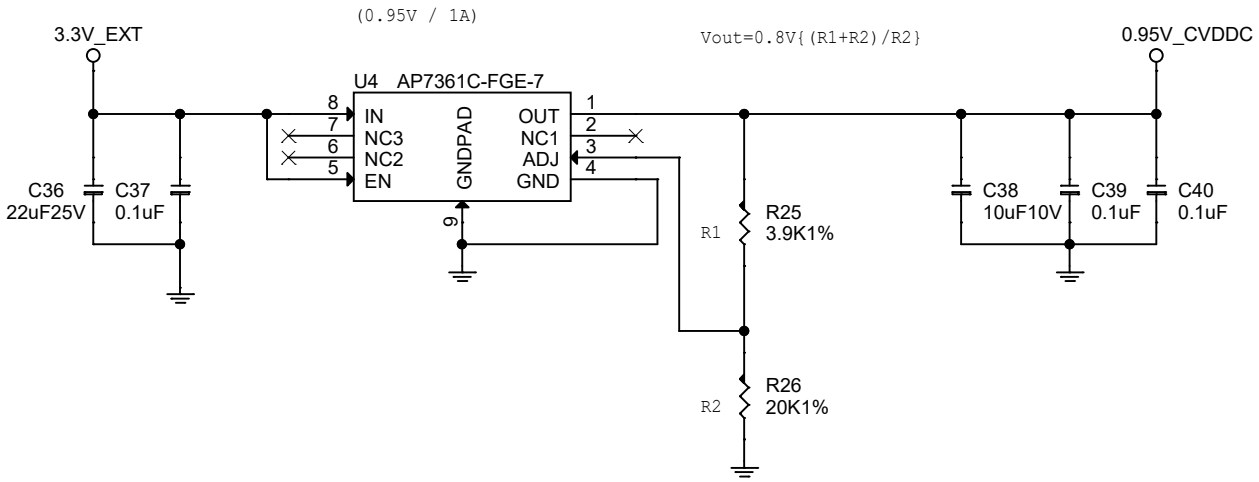
Capacitor y tiempo de arranque suave							
Css (nF)	ABIERTO	3.3	4.7	10	15	22	33
Tiempo SS (ms)	1.2	2.1	2.5	3.8	5.1	7.0	9.8

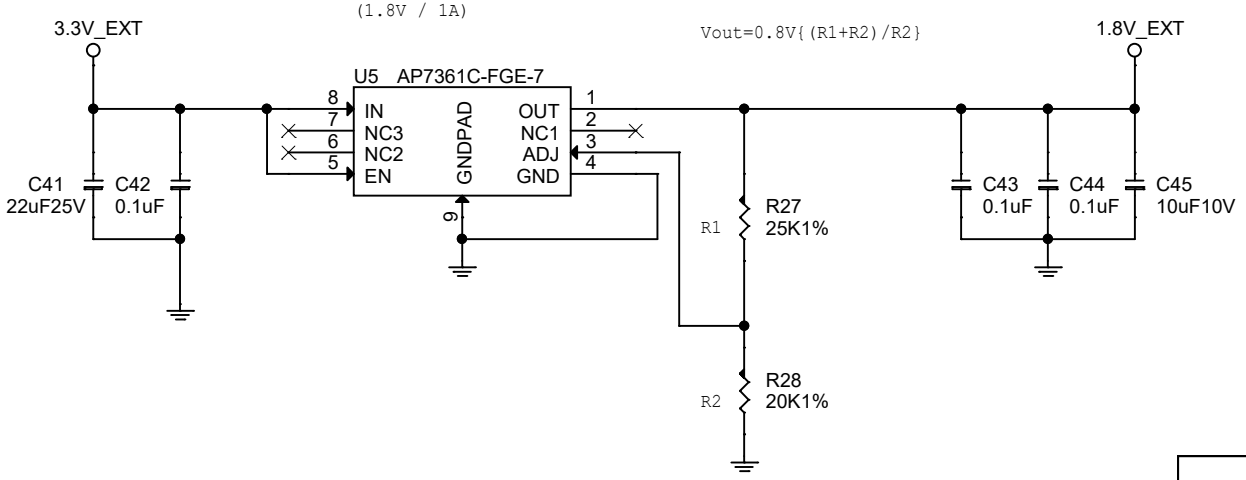
Configuración del módulo

Vout = [(1.43 / Rset) + 1] x 0.4

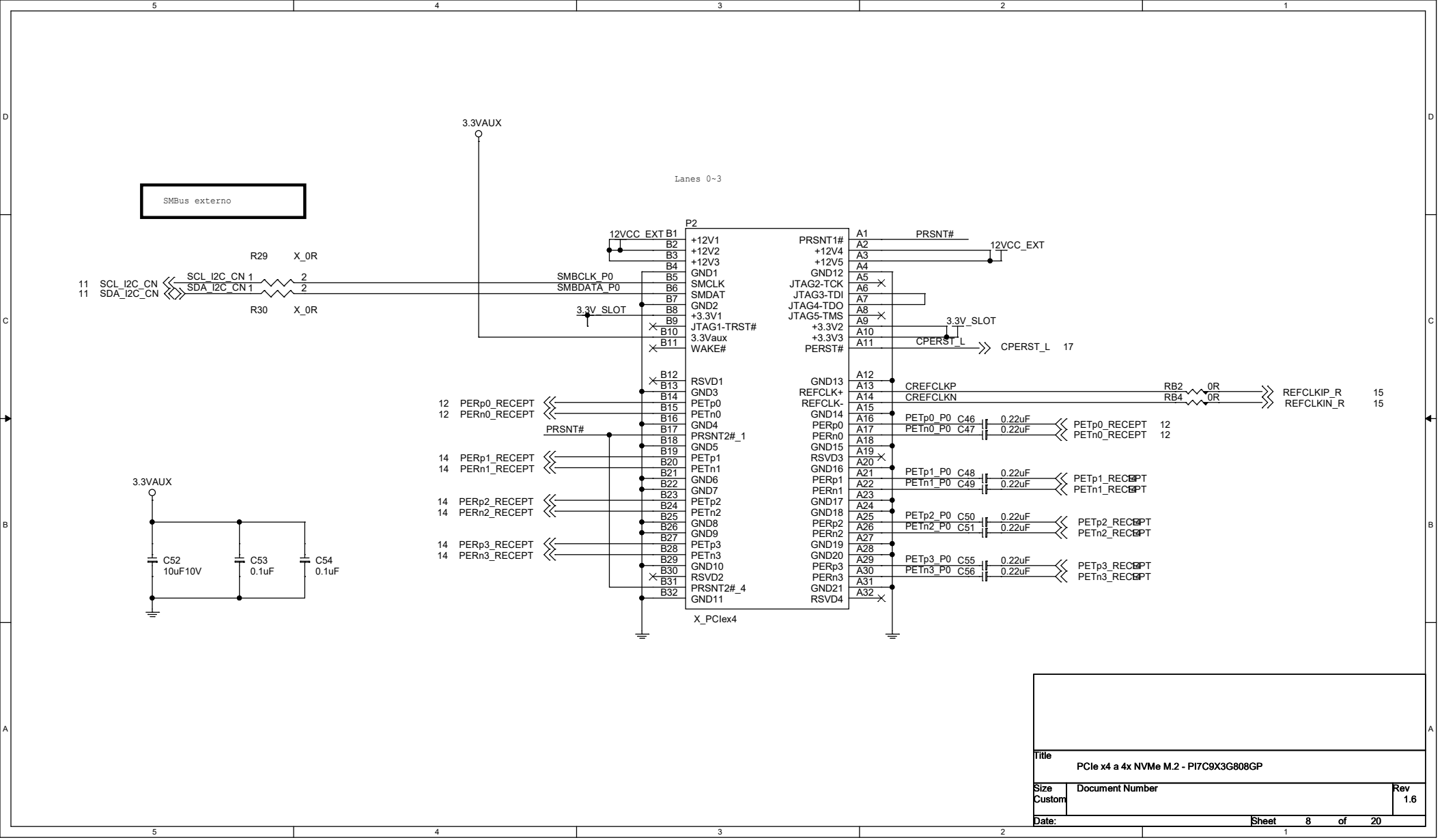
Vout	Rset	R(RT)	fsw
0.95V	2.45K	1M	250KHz

File:	PCIe x4 a 4x NVMe M.2 - PSC3XG3208GP	
Size: Custom	Document Number	Rev: 1.6
Date:	Sheet 8 of 39	



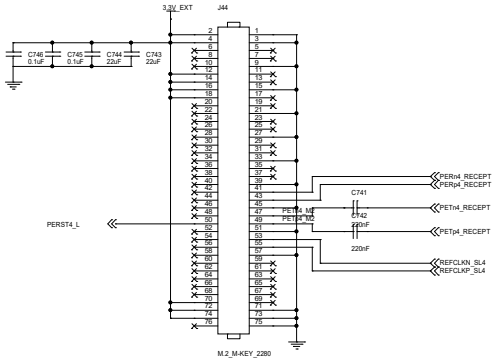


Title: PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: A	Document Number	Rev: 1.6
Date:	Sheet 7 of 20	

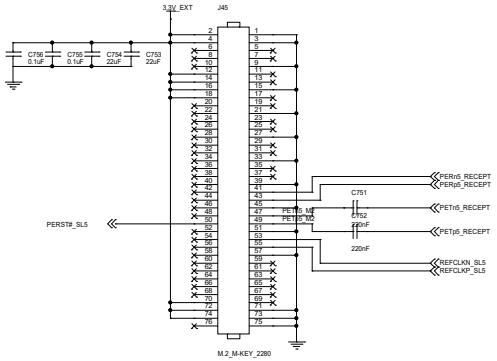


Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number	Rev: 1.6
Date:	Sheet 8 of 20	

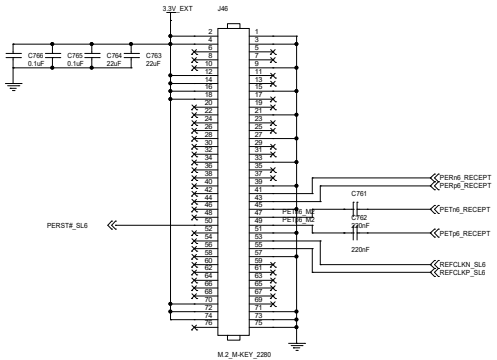
J44 - SSD 1: puerto P4 x1 (lane 4)



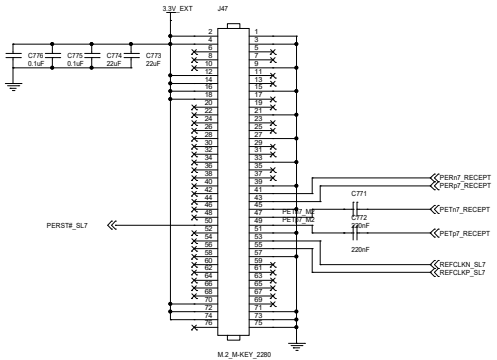
J45 - SSD 2: puerto P5 x1 (lane 5)



J46 - SSD 3: puerto P6 x1 (lane 6)



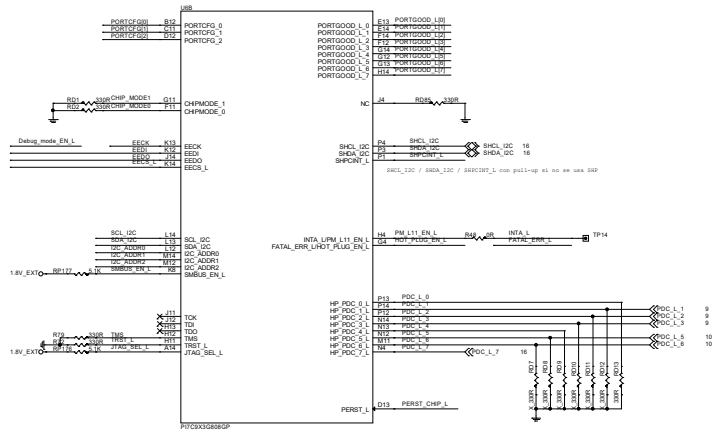
J47 - SSD 4: puerto P7 x1 (lane 7)

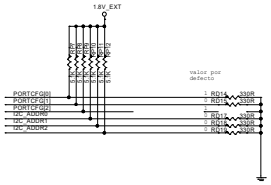


M.2 M-key 2280 (NVMe x4): 3.3V_EXT, caps AC 220 nF en TX, REFCLK04-7 (55/53), PERST# (50)

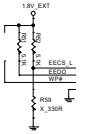
Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number	Rev: 1.6
Date:	Sheet 10 of 20	

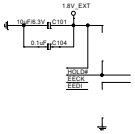
CHIP_MODE[1:0]	Modo
00	Normal
01	Reservado
10	Reservado
11	Reservado

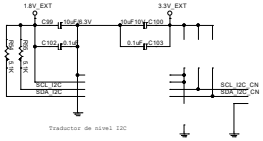




PORTCFG[2]	PORTCFG[1]	PORTCFG[0]	Modo de funcionamiento
0	0	0	Reservado
0	0	1	2 puertos - 8 lanes
0	1	0	3 puertos - 8 lanes
0	1	1	4 puertos - 8 lanes
1	0	0	5 puertos - 8 lanes
1	0	1	8 puertos - 8 lanes
1	1	0	Reservado
1	1	1	Reservado



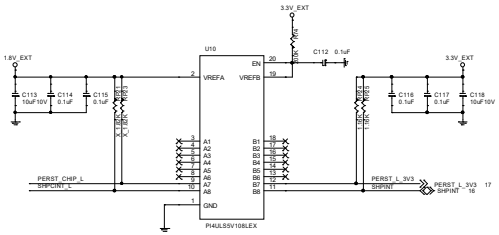


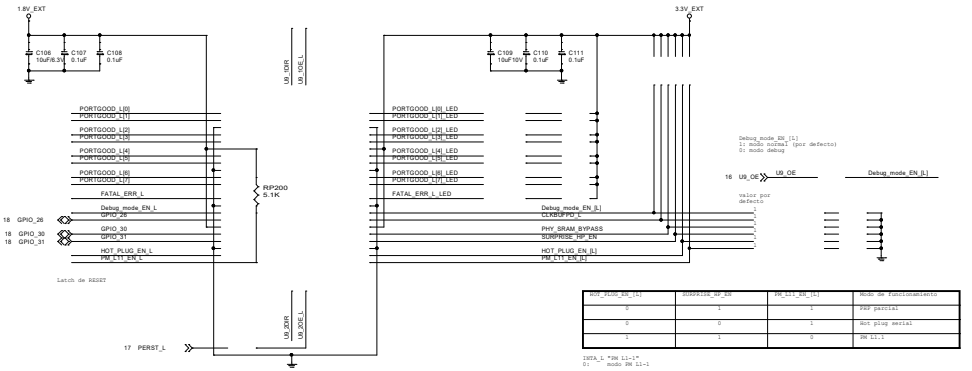


8 SCL_I2C_CN
8 SDA_I2C_CN

Traductor de nivel I2C

Traductor de nivel para PERST / SHP





HOT_PLUG_EN_[L]	SURPRISE_HP_EN	PM_L11_EN_[L]	Modo de funcionamiento
0	1	1	PHP parcial
0	0	1	Hot plug serial
1	1	0	PM L1.1

INTA_L "PM L1-1"
0: modo PM L1-1

FATAL_ERR_L "HOT_PLUG_EN_L"
0: GPIO[31:0] modo hot-plug paralelo

ENTRADAS DE CONTROL		CIRCUITOS DE SALIDA		
OE#	DIR	PUERTO A	PUERTO B	OPERACIÓN
L	L	Habilitado	Hi-Z	A-> A
L	H	Hi-Z	Habilitado	A-> B
H	X	Hi-Z	Hi-Z	Aislado

GPIO[26] "CSWGPIO_L"
1: buffer de reloj habilitado
0: buffer de reloj deshabilitado

GPIO[30] "PHY_SRAM_BYPASS"
1: bypass de la SRAM (por defecto)
0: sin bypass de la SRAM

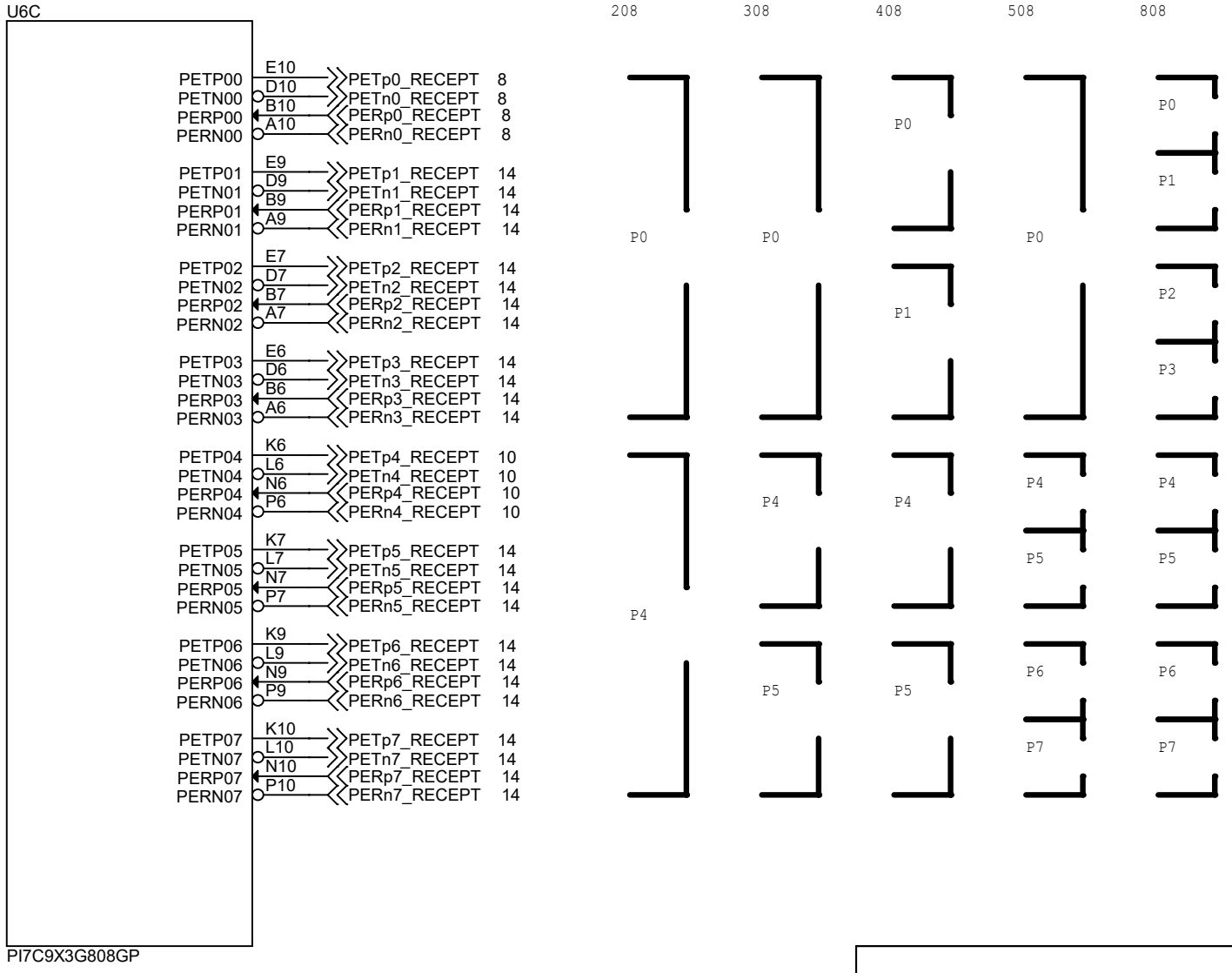
GPIO[31] "SURPRISE_HP_EN"
1: modo normal para el manejo de LED
PHP parcial (sin expansor IO ni CPLD)
0: hot-plug administrado, PCIe SHP
PCIe PHP (sin expansor IO ni CPLD)

Entrada

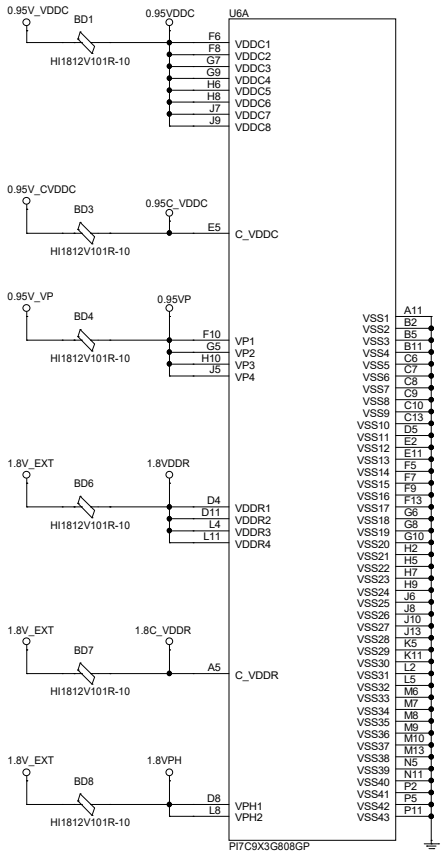


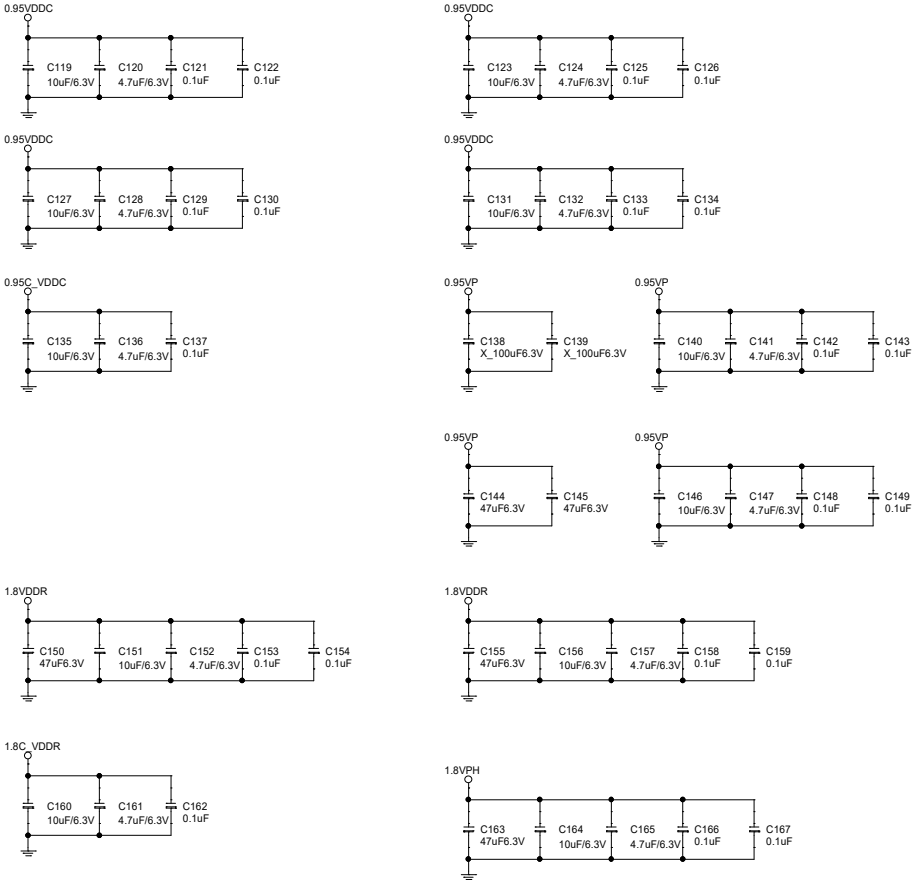
Los pines de control se alimentan de VCCA

File	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: Custom	Document Number		Rev 1.0
Date:	Sheet 11 of 20		

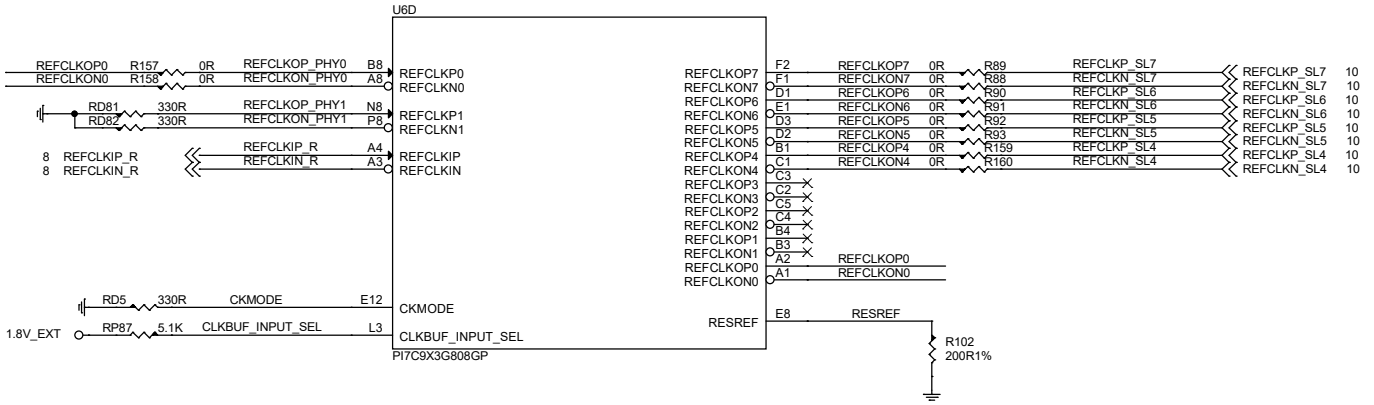


Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size A	Document Number	Rev 1.6
Date:	Sheet 12 of 20	

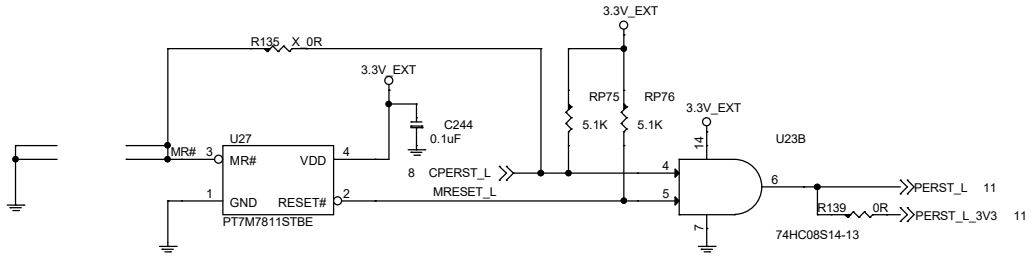


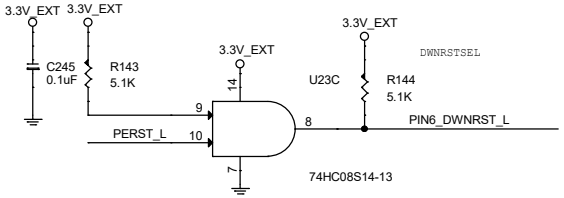


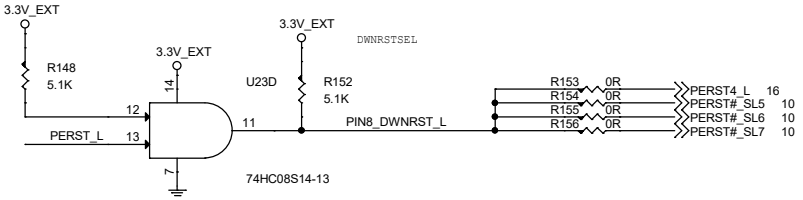
Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number	Rev: 1.6
Date:	Sheet 13 of 20	



Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size B	Document Number	Rev 1.6
Date:	Sheet 15 of 20	







Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size B	Document Number	Rev 1.6
Date:	Sheet 17 of 20	